MILITARY SPECIFICATION SHEET

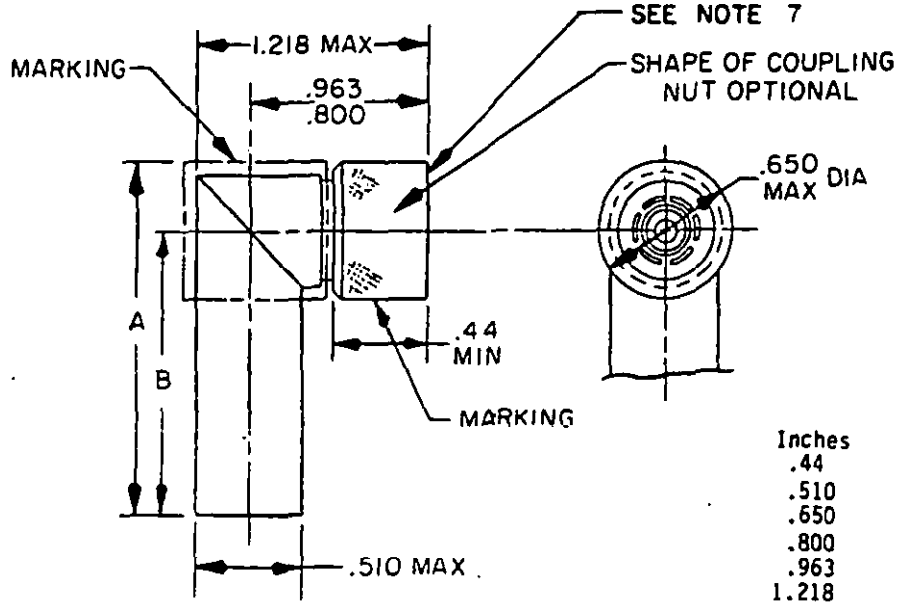
CONNECTORS, PLUG, ELECTRICAL, COAXIAL, RADIO FREQUENCY,

(SERIES TNC (CABLED), PIN CONTACT, RIGHT ANGLE, CLASS 2)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-39012.

MARKING IMPLEMENTATION DATE, CATEGORY B, SEE TABLE IV



Inches	mm
.44	11.18
.510	12.95
.650	16.51
.800	20.32
.963	24.46
1.218	30.94

NOTES:
1. Dimensions are in inches.
2. For dimensions A and B, see table I and III.
3. Metric equivalents are given for general information only.
4. Wrench flats are to accommodate standard wrench openings in accordance with FED-STD-H25, appendix 10.
5. Dimension A defines the maximum length of the connector when assembled to the appropriate cable.
6. All undimensioned pictorial configurations are for reference purposes only.
7. Series TNC, pin contact interface in accordance with MIL-STD-348.

FIGURE 1. General configuration.

AMSC N/A FSC 5935

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TABLE I. Dash numbers, cross reference, and dimensions.

Dash number 1/	Applicable cable M17/#	Dimensions	Inches (millimeters) Maximum
Category A - Field serviceable (no special tools required)			
0101 3/	028-RG058 084-RG223* 128-RG400 060-RG142@		
0102 3/	29-RG59*Δ 030-RG062Δ 90-RG71Δ 97-RG210Δ	A	1.750 (44.45)
0103 3/	054-RG122*		
0104 3/	111-RG303@*		
0117 3/	110-RG302*@Δ		
0118 3/	113-RG316*@ 119-RG174		
Category C - Field replaceable (MIL-C-22520 crimp tool) 2/ See footnote symbol next to applicable cable for crimp die			
0010 3/	028-RG058*- 111-RG303@-		
0011 3/	084-RG223*- 128-RG400- 060-RG1420-		
0012 3/	29-RG59*Δ>& 030-RG062Δ>& 97-RG210Δ>&	A	2.000 (50.80)
0013 3/	054-RG122*&		
0014 3/	90-RG71*Δ>&		
0021 3/	110-RG302*@>Δ&		
0022 3/	113-RG316@*Σ 119-RG174Σ		
Category D - Field replaceable - Defined piece parts 2/ 4/ 5/			
0501 3/	054-RG122*		
0502 3/	095-RG180*Δ	A	1.859 (47.23)
0503 3/	060-RG142* 128-RG400	B	1.468 (37.31)
0504 3/	028-RG058		

See footnotes on the following page.

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1/ For cross reference of dash number to superseded part number or type designation, see table IV.

2/ These connectors are assembled, using the applicable crimp tool, to the specified cables stripped as shown on figure 3.

3/ These connectors have captivated center contacts.

4/ Complete connector assembly shall consist of a body, center contact, ferrule, and assembly instructions.

5/ Not to be used in Army equipment.

The latest version of each cable shall be applicable.

* Cable to be used when performing tests requiring cable except as in notes @ and Δ.

@ Cable to used for the +200°C temperature cycling tests.

Δ These are not 50 ohm cables; therefore, when attached to the specified connectors, VSWR, RF leakage, and insertion loss are not applicable.

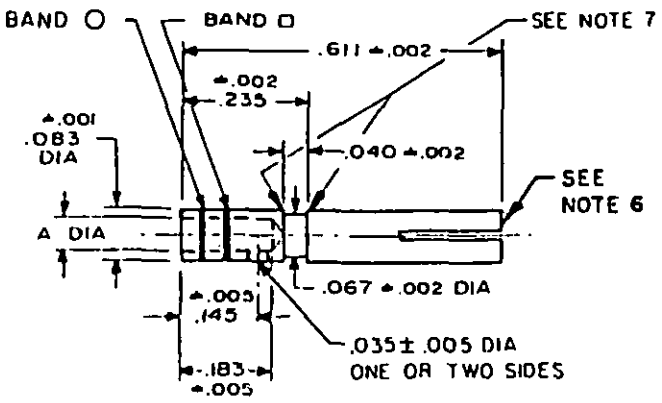
Σ M22520/5-35 Closure B or M22520/5-03 Closure A

& M22520/5-41 Closure B or M22520/5-05 Closure B
M22520/5-09 Closure A

- M22520/5-19 Closure B or M22520/5-05 Closure A
M22520/5-11 Closure A
M22520/5-57 Closure A

> M22520/5-19 Closure A or M22520/5-07 Closure A
M22520/5-13 Closure A
M22520/5-59 Closure A

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CENTER CONTACT

Inches	mm
.001	0.03
.002	0.05
.005	0.13
.015	0.38
.017	0.43
.033	0.84
.035	0.89
.040	1.02
.043	1.09
.067	1.70
.083	2.11
.145	3.68
.175	4.45
.183	4.65
.206	5.23
.212	5.38
.220	5.59
.235	5.97
.245	6.22
.250	6.35
.256	6.50
.261	6.63
.263	6.68
.297	7.54
.500	12.70
.611	15.52

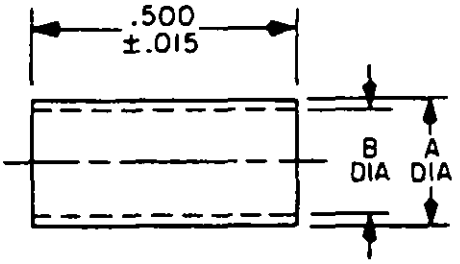
Dash no.	Contact no. 2/	A +.001 -.002	Basic crimp tool 1/	Crimp die or positioner	Crimp tensile min	Color band □	Color band ○
0501	30-12	.033			10 lbs	Orn	Violet
0502	30-11	.017	M22520/1-01	M22520/1-12	6 lbs	Blue	
0503 0504	30-10	.043	M22520/1-01	M22520/1-12	20 lbs	Red	

1/ Class 2 tool may be used by OEM (see MIL-C-22520).
2/ Contact numbers and ferrule numbers are for identification only.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Contact material shall be phosphor bronze or copper beryllium.
4. Crimp tensile test shall be in accordance with MIL-C-39029.
5. Copyright notice: All information disclosed in these specification sheets which is or may be copyrighted is reproduced herein with the express permission of the copyright owner.
6. Form socket to mate with .052/.054 (1.32/1.37 mm) diameter and .195 (4.95 mm) depth pin.
7. .003 (0.08 mm) maximum break.

FIGURE 2. Contact and ferrule dimensions for category D only.

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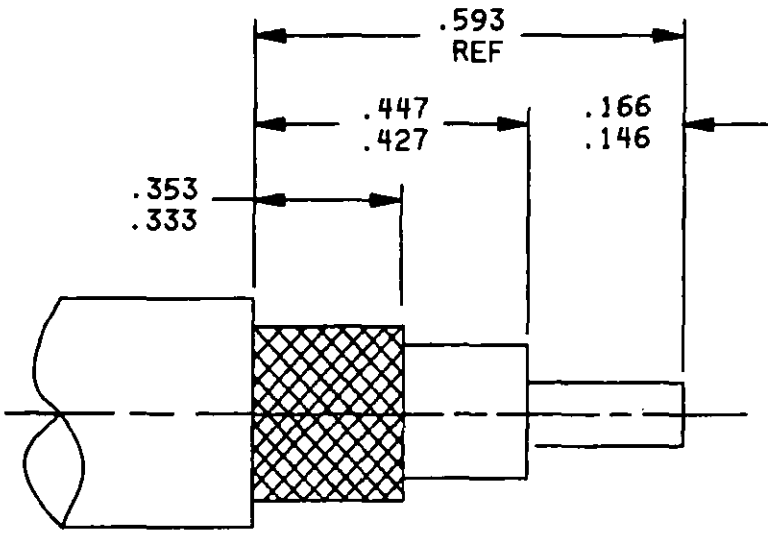
CRIMP FERRULE

Dash no.	Ferrule no. 2/	A ±.001	B ±.001	Basic crimp tool 1/	Crimp die or positioner M22520/5-
0501 0502	30-50	.212	.175	M22520/5-01	05, 41 Closure B or 9 Closure A
0503	30-51	.250	.220		05, 11, 57 Closure A or 19 Closure B
0504	30-52	.245	.206		

1/ Class 2 tool may be used by OEM (see MIL-C-22520).
2/ Contact numbers and ferrule numbers are for identification only.

FIGURE 2. Contact and ferrule dimensions for category D only - Continued.

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Inches	mm
.146	3.71
.166	4.22
.333	8.46
.353	8.97
.427	10.85
.447	11.35
.593	15.06

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 3. Cable stripping dimensions for field replaceable connectors.

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ENGINEERING DATA:

Nominal impedance: 50 ohms.

Frequency range: 0 to 11,000 MHz.

Voltage rating:

500 volts rms, maximum working voltage at sea level.

125 volts rms, maximum at 70,000 feet.

Temperature rating: -65°C to +165°C.

REQUIREMENTS:

Dimensions and construction: See figure 1.

Force to engage and disengage:

Longitudinal force: Not applicable.

Torque: 2 inch-pounds, maximum.

Coupling proof torque: 15 inch-pounds, minimum.

Inspection conditions: Coupling torque: 4 to 6 inch-pounds.

Mating characteristics: Reference MIL-STD-348 and figure 2 for dimensions.

Outer contact:

Test ring ID: .319 maximum, 16 microinch finish.

Insertion force: 5 pounds, maximum when inserted a minimum of .093.

Contacts with slotted members: Shall contact a .324, minimum diameter ring within .031 of their tip ends.

Hermetic seal: Not applicable.

Leakage (pressurized connectors): Not applicable.

Insulation resistance: Method 302 of MIL-STD-202, test condition B, 5,000 megohms, minimum.

Center contact retention: 6 pounds, minimum axial force. Applicable to captivated-center-contact connectors only.

Corrosion (salt spray): Method 101 of MIL-STD-202, test condition B.

Voltage standing wave ratio (VSWR): From 500 to 11,000 MHz, or approximately 80 percent of upper cutoff frequency of the cable, whichever is lower; 1.30, maximum.

Swept frequency VSWR test setup:

Item 6: VSWR shall be less than 1.02+ .003 F (F in GHz).

Item 16: VSWR shall be less than 1.02+.003 F (F in GHz).

Second step of VSWR checkout procedure - VSWR shall be less than 1.06+.007 F (F in GHz).

Group B inspection: VSWR shall be less than 1.05+.005 F (F in GHz).

Qualification and group C inspection: VSWR shall not exceed 1.10.

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Connector durability: 500 cycles, minimum at 12 cycles per minute, maximum. The connector shall meet the mating characteristics and force to engage and disengage requirements.

Contact resistance: In milliohms, maximum.

	Initial	After environment
Center contact	2.0	2.5
Outer contact	.2	Not applicable
Braid to body	.1	Not applicable

Dielectric withstanding voltage: Method 301 of MIL-STD-202, 1,500 volts rms at sea level.

Vibration, high frequency: Method 204 of MIL-STD-202, test condition B. No discontinuity permitted.

Shock: Method 213 of MIL-STD-202, test condition I. No discontinuity permitted.

Thermal shock: Method 107 of MIL-STD-202, test condition B, except test high temperature shall be +85°C. High temperature shall be +200°C for connectors using 200°C cables (see tables I and III).

Moisture resistance: Method 106 of MIL-STD-202. No measurements at high humidity. Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity.

Corona level:

Voltage: 375 volts rms, minimum.

Altitude: 70,000 feet.

RF high potential withstanding voltage:

Voltage and frequency: 1,000 volts rms tested at a frequency from 5 to 7.5 MHz.

Leakage current - Not applicable.

Cable retention force:

Noncrimp assemblies: 40 pounds, minimum.

Crimp assemblies:

10 pounds, minimum for cables .155 - .189 OD.

20 pounds, minimum for cables .190 - .229 OD.

30 pounds, minimum for cables .230 - .249 OD.

40 pounds, minimum for cables .250 OD and larger.

Coupling mechanism retention force: 100 pounds, minimum.

RF leakage: -60 dB minimum, tested at a frequency between 2 and 3 GHz.

Insertion loss:

.21 dB maximum at 9 GHz.

$.07 \sqrt{F\ (GHz)}$ dB maximum tested at 3 GHz and 6 GHz.

Part number: M39012/30 (dash number from table I or 'B' number from table III).

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TABLE II. Group qualification.

Group	Submission and qualification of any of the following connectors	Qualifies the following connectors
	M39012/30	M39012/30
I	0101 0103 0104 0118	0101 0102 0103 0104 0117 0118
II	0102 0117	0102 0117
III	B0005 B0006 B0008 B0015 B0016 B0020	B0005 B0006 B0007 B0008 B0009 B0015 B0016 B0019 B0020
IV	B0007 B0009 B0019	B0007 B0009 B0019
V	0010 0011 0013 0022	0010 0011 0012 0013 0014 0021 0022
VI	0012 0014 0021	0012 0014 0021
VII	0501 0503 0504	0501 0502 0503 0504
VIII	0502	0502

NOTE: If a connector manufacturer produces a connector which meets all the requirements for two or more connector part numbers (within the same series), the manufacturer may receive qualification approval for two or more connector part numbers qualifying the one connector. It is not necessary that such connectors be in the same group. Each connector, however, must be marked with its own appropriate part number. For group qualification, the connectors must be of similar design.

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TABLE III. Category B - Nonfield replaceable (special tools may be required).

Not for Air Force or Navy use. For OEM use only.

Dash number 1/ 2/ 3/	Applicable cable M17/#	Dimensions	Inches (millimeters) maximum
M39012/30900005	028-RG058*	A	2.000 (50.80)
M39012/30800006	060-RG142∂* 128-RG400		
M39012/30800007	29-RG59*∆ 030-RG062∆ 97-RG210∆		
M39012/30800008	054-RG122*		
M39012/30800009	90-RG71*∆		
M39012/30800015	111-RG303*@		
M39012/30800016	084-RG223*		
M39012/30800019	110-RG302∂*∆		
M39012/30800020	113-RG316∂* 119-RG174		

1/ For cross reference of dash number to superseded part number or type designation, see table IV.

2/ For maintenance replacements for category B, see table V.

3/ Inactive for new design.

The latest version of each cable shall be applicable.

* Cable to be used when performing tests requiring cable except as in notes @ and ∆.

∂ Cable to used for the +200°C temperature cycling tests.

∆ These are not 50 ohm cables; therefore, when attached to the specified connectors, VSWR, RF leakage, and insertion loss are not applicable.

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TABLE IV. Cross reference of part numbers.

Preferred part number	Substitute for part number or type designation 1/
M39012/30-	M39012/30-
B0005	0005
B0006	0006
B0007	0007
B0008	0008
B0009	0009
B0015	0015
B0016	0016
B0019	0019
B0020	0020
0101	0001
0102	0002
0103	0003
0104	0004
0117	0017
0118	0018

1/ The superseded part number or the type designation is for cross reference only. Where a superseded part number or type designation is not given, none was assigned or will be assigned. The part number M39012/30-XXXX shall be used in all cases for marking and identifying the connector.

2/ The new "B" part numbers will be required marking 6 months after the date of this specification. The previous part number may be used in the interim.

TABLE V. Maintenance replacements for category B.

Category B number* Inactive for new design	Category C dash number	Category A dash number	Category D dash number
B0005	0010	0101	0504
B0006	0011	0101	0503
B0007	0012	0102	---
B0008	0013	0103	0501
B0009	0014	0102	---
B0015	0010	0114	---
B0016	0011	0101	---
B0019	0021	0117	---
B0020	0022	0118	---

*Category B connectors are for original installation only. They will not be stocked or acquired by the Government.

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Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - CR, EA, MI
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AT, AV
Navy - AS, MC, OS, SH
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

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